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Controlled swelling behavior and stable cycling of silicon/graphite granular composite for high energy density in lithium ion batteries

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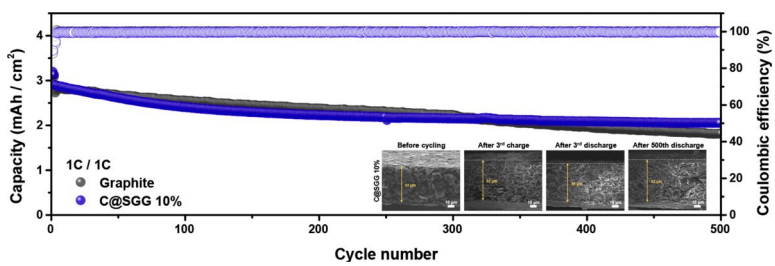
HIGHLIGHTS

- C@SGG was developed by facile and scalable processes.
- The C@SGG 10% electrode showed the long-term cycling performance over 500 cycles.
- The electrode swelling behavior was measured by in-situ thickness measurement.
- The C@SGG 10% electrode exhibited stable electrode swelling behaviors.

GRAPHICAL ABSTRACT



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ABSTRACT

The severe swelling behavior of silicon-based electrodes caused by the large volume expansion of Si associated with Li still prevents the practical implementation of Si. Here, we report the amorphous carbon coated silicon/graphite granule (C@SGG) as an attractive anode material to control the swelling behavior and achieve improved electrochemical properties. With the structural uniqueness including inner buffer space, large contact area of silicon and graphite, and the uniform surface carbon coating layer, the C@SGG electrode exhibits stable cycling performance and regulated electrode swelling behavior. C@SGG shows high reversible capacity of 1150 mA h g^{-1} and improved initial Coulombic efficiency of 85%. We evaluate C@SGG from a practical point of view through the characterization of pouch full cells prepared with $LiCoO_2$ as a cathode with a high areal capacity of 3.4 mA h cm^{-2}. The electrode prepared with the C@SGG of 10% and graphite of 90% (C@SGG 10% electrode) exhibits a stable cycling performance with a capacity retention of 71% over 500 cycles at 1 C-rate. Furthermore, the C@SGG 10% electrode exhibits stable electrode swelling behaviors comparable to the graphite electrode, indicating an initial expansion of 16.0% at a charged state and a cycling expansion of 6.7% over 50 cycles.

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1. Introduction

Lithium-ion batteries (LIBs) have been widely used in most of electronic devices and become inseparable things in our lives. However, conventional LIBs already reached the limit in terms of the energy density of 250 Wh/kg [1–4]. To secure sufficient mileage of electric vehicles (EVs), it is necessary to develop the LIBs with high energy density. Silicon has been widely studied for the sake of its high theoretical specific capacity of 3572 mA h g^{-1} and 8322 mA h cm^{-3} ($Li_{15}Si_4$), and a relatively low working voltage (~0.5 V vs Li/Li^+) [5,6]. However, Si experiences the large volume changes up to 300% associated with Li, which leads to poor electrochemical properties caused by loss of the electrical contact between active materials and nagging electrolyte decomposition with a continuous formation of a SEI (solid electrolyte interphase) layer [7,8].

Various Si nanostructures have been explored to overcome the innate disadvantages of Si by mitigating the stress caused by volume changes during cycling [9,10]. However, nanoscale materials have the limitation in the increase of the volumetric energy density due to the low bulk density and poor Coulombic efficiency due to the large surface area for practical applications. In this regard, microscale materials, especially Si and carbon (Si/C) composite, have been significantly studied as a practical point of view. Microscale Si/C composite materials with the void space inside the composite exhibited the improved cycle performances due to the carbon material and inner void space which can act as a buffer for the volume change of Si [11–13]. However, the Si/C composite electrode with a high areal capacity level (>3 mA h cm^{-2}) have not been sufficiently studied. With increase of the areal capacity of the Si/C composite electrode, their electrochemical properties are dramatically degraded due to the increased diffusion lengths of charge carriers in thicker electrode [14]. Furthermore, more severe volume expansion of the composite electrode causes the electrode swelling issues [15,16]. The battery safety is of paramount importance for applications that have consumers directly interacting with the device especially in consumer electronics and EVs [17,18]. Therefore, it is necessary to understand the swelling behavior as well as the electrochemical properties of the Si/C composite based electrode with the high areal capacity.

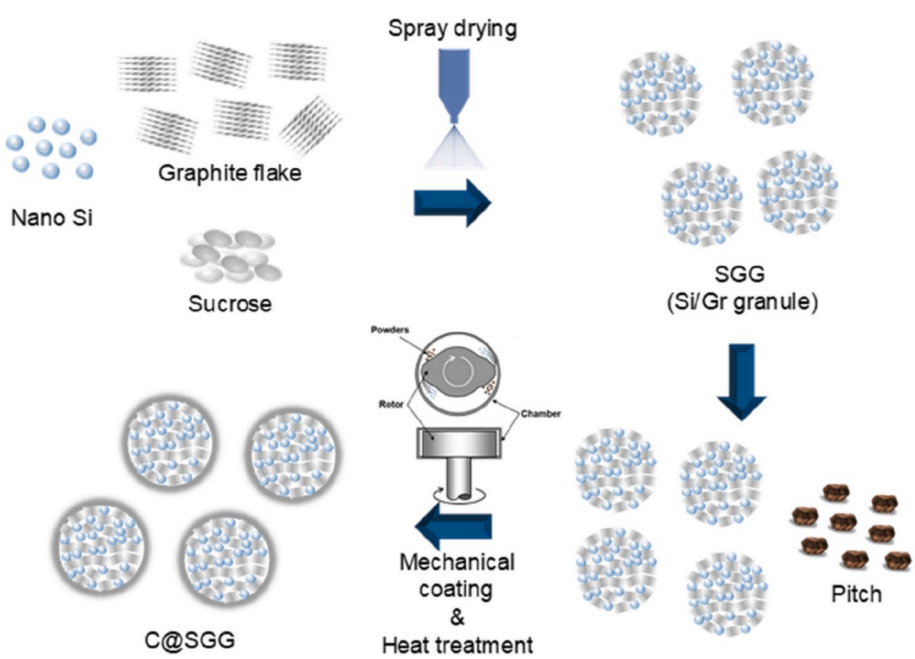
Here, we report the silicon/graphite granular composite with the uniform amorphous carbon coating layer (C@SGG) as an anode material to improve the energy density and control the electrode swelling behavior of LIBs. C@SGG was prepared via facile and low-cost processing route where silicon/graphite granule (SGG) was synthesized through the spray drying and subsequently, the amorphous carbon was uniformly coated on the surface of SGG through the mechanical coating method (Scheme 1). The homogeneously distributed Si nanoparticles and graphite flakes inside the particle could improve the kinetics of Li-ion with the large contact area. Furthermore, the uniform inner free space and outer amorphous carbon coating layer could buffer the volume change of Si, which enables the controlled swelling behavior of the C@SGG electrode during cycling. With those advantages, The C@SGG electrode exhibited the stable cycling performance and improved initial Coulombic efficiency of 85% compared to those of the SGG electrode. To explore the possibility of the practical implementation of the C@SGG active material, the electrochemical properties of the C@SGG and graphite composite electrode with a high areal capacity of 3.4 mA h/cm^2 were evaluated using $LiCoO_2$ (LCO) as a cathode. The electrode composed of C@SGG of 10% and graphite of 90% (C@SGG 10% electrode) exhibited the long-term cycling performance with a capacity retention of 83% over 100 cycles and 71% over 500 cycles under 1C/1C (charging/discharging) rate in the pouch full cell evaluation. Furthermore, the C@SGG 10% electrode showed the outstanding discharge rate capabilities up to 3C rate with lower interfacial resistances and higher Li-ion diffusion coefficient compared to those of the graphite electrode. The swelling behavior of the electrodes were monitored through in-situ electrode thickness measurement system. The C@SGG 10% electrode exhibited stable electrode swelling behavior over 50 cycles, showing initial expansion of 16.0% at charged state and cycling expansion rate of 6.7%.

2. Experimental

2.1. Materials synthesis

2.1.1. Preparation of SGG

Before spray drying, the graphite flake was prepared through the planetary ball milling (FRITSCH, GERMANY), as shown in Fig. S1. The weight ratio of graphite flake, Si (HANBYUL), and sucrose (Aldrich) in the raw materials for the synthesis of SGG was designed as 55:35:10 and they were dispersed in deionized water. After magnetic stirring for 2 h, the solution was spray dried through the spray dryer (EYELA, JAPAN) to



Scheme 1. Schematic illustration of the preparation procedure for SGG by spray drying and C@SGG by mechanical coating method.

synthesize SGG with porous structure under the following conditions: the rotating speed of the atomizer was 500 rpm; atomizing 190 kPa; The blowing speed 0.55 m^2/min; the pumping speed was 200 mL/h; the inlet temperature of the spray dryer was 110 °C.

2.1.2. Preparation of C@SGG

To form the uniform surface coating layer on SGG, mechanical coating method by using attrition type mill was employed [19]. Initially, the temperature of the batch was heated to 90 °C using a heater, and then SGG and pitch (coal-tar pitch, TCK) were put into the weight ratio of 8:2. The input amount of pitch was determined based on the TGA result (Fig. S2). The pitch was uniformly coated on the surface of SGG at 500 rpm for 30 min by the attrition type mill [20]. The as prepared powders were heated at 800 °C for 2 h in a quartz tube furnace under an Ar atmosphere to form the amorphous carbon coating layer on SGG. The resulting composite, C@SGG, was ground for the fabrication of the electrode.

2.2. Material characterization

For analysis of the morphology of the powder and electrodes, field-emission scanning electron microscopy (FE-SEM, JEOL JSM07600F) was employed. The electrode after battery test was washed using dimethyl carbonate (DMC, Aldrich) as a solvent before their SEM analysis. X-ray diffraction (XRD, Rigaku SmartLab) was used to analyze SGG and C@SGG. X-ray photoelectron spectrometry (XPS, VG microtech ESCA2000) was used for analysis of SGG and C@SGG and the binding energy was calibrated based on the peak of C 1s to 284.8 eV. To verify the surface coating layer and the elemental mapping of Si and C in the particle, transmission electron microscopy (TEM, JEOL JEM-2100F) was used. Si content of SGG and C@SGG was identified through ICP-OES (Thermo Fisher Scientific). Thermogravimetric analysis (TGA) was carried out on a TA Instruments Q500 in successive steps with the same specimen to verify the remaining weight of pitch. The weight loss was recorded upon heating to 800 °C at a rate of 20 °C min^{-1}. Raman spectroscopy was measured by LabRam Aramis (Horiba Jobin Yvon) and An Ar ion laser was used as an excitation source of 514 nm. Nitrogen gas adsorption–desorption isotherm analysis was performed on the graphite, SGG, and C@SGG samples at −196 °C using 3Fex (Micromeritics). Before test, all samples were dried at 80 °C for 24 h under vacuum. The specific surface area was calculated by Brunauer–Emmett-Teller (BET) equation.

2.3. Electrochemical measurements

For slurry preparation, the active material, conducting carbon (Ketjen black), carboxymethyl cellulose (CMC, Aldrich), and styrene butadiene rubber (SBR, 40% emulsion) were placed into the water at a weight ratio of 97:0.5:1:1.5 based on the solid, and then mixed with a thinky mixer at 2000 rpm for 5 min. Then, the prepared slurry was coated onto a copper current collector and dried at 110 °C for 10 min. Before fabrication of battery, all electrodes were dried at 120 °C for 4 h in a vacuum oven. For the half-cell test, 2032 round type coin cells were fabricated using lithium foil with a thickness of 200 μm as a counter electrode. For the full cell test, the anode was pressed by roll press machine to achieve a high energy density, and pouch cells were fabricated using $LiCoO_2$ (LCO) with a high areal capacity of 3.4 mA h cm^{-2} (21 mg of LCO cm^{-2}) as a cathode where N/P ratio was 1.08. The formation of the pouch full cells were 0.1C and 0.2 C charging of CC-CV and discharging of CC modes, respectively between 4.2 and 2.8 V. 1.0 M $LiPF_6$ in ethylene carbonate/dimethyl carbonate/diethyl carbonate with fluoroethylene carbonate additives (EC/DMC/DEC (1/1/2 vol%) with 10 wt% FEC, PANAX StarLyte) was used as electrolyte, and a polypropylene (PP) separator was used. All of the cells were tested using a battery cycle tester (TOSCAT 3000, Toyo System, Tokyo, Japan). EIS was measured at OCV (open circuit voltage) of each cell with a frequency from 1 to 100,000 Hz and an amplitude of 10 mV using a potentiostat (Autolab, PGSTAT – 72637). The Li ion diffusion coefficients could be calculated by the equation of $D_{Li^+} = \frac{R^2T^2}{2A^2n^4F^4C^2\sigma^2}$ where D_{Li^+} is Li diffusion coefficient, R is the gas constant (8.314 J K^{-1} mol^{-1}), T is the absolute temperature of 298.15 K, A is the electrode surface area (electrode area × surface area of active materials), n is the number of transferred electrons, F is the Faraday constant (96500 C mol^{-1}), C is the concentration of Li-ions, and σ is the Warburg factor (the slope of the Warburg plot). The in-situ thickness change of the pouch full cell was observed using a linear gauge (LGK-0110, Mitutoyo, Japan) during cycling.

3. Results and discussion

3.1. Material characterization

Fig. 1 shows the microstructural images of SGG and C@SGG particles. The SGG particle, prepared by the spray drying, shows a granular structure with the homogeneous distribution of Si nanoparticles and graphite flakes, as shown in Fig. 1a and Fig. S3. The as prepared SGG has a large surface area and poor mechanical property due to the weak binding strength attributed to the hydrophilic surface of Si nanoparticles and the hydrophobic surface of graphite, which leads to the poor initial Coulombic efficiency and cycle performance [21,22]. The amorphous carbon layer was coated on the surface of SGG particle to overcome these problems. The pitch was conformally coated onto the surface of SGG particles through the mechanical coating process in a N_2 atmosphere, and then the pitch coated SGG particles were heated at 800 °C for 2 h in an Ar atmosphere for the formation of the amorphous carbon coating layer [23]. To characterize the carbon materials Raman analysis was employed, as shown in Fig. S4. The pristine graphite is highly oriented with a very low I_D/I_G ratio of 0.08, however graphite flake shows a higher I_D/I_G ratio of 0.55 due to the defect by the mechanical breakage [24]. Raman spectrum of C@SGG exhibits typical amorphous carbon with a I_D/I_G ratio of 0.92 [25]. The whole process is a facile, low cost, and scalable approach. The uniform amorphous carbon coating layer with a thickness of 30 nm is clearly observed, as shown in Fig. 1b and c. The specific surface area of C@SGG (~1.70 m^2 g^{-1}) is much lower than that of SGG (~96.18 m^2 g^{-1}), which proves the uniform amorphous carbon coating layer of C@SGG (Fig. S5). The SEM EDS-mapping images and the SEM image in backscattered electron mode clearly exhibit the uniform distribution of Si and graphite flake (Fig. 1d–f and Figs. S6 and S7). Fig. 2a shows the cross-sectional SEM image of the C@SGG particle, which clearly confirms the granular structure of C@SGG particle with the inner void space. These free space in the C@SGG particle enables the facile relaxation of the stress caused by the volume change of Si during cycling, which can improve the electrochemical performances of the C@SGG without the fracture of the particles. The XRD diffraction patterns for SGG and C@SGG powders (Fig. 2 d) are well matched with the peaks of graphite corresponding to (002), (100), (101), and (004), and silicon corresponding to (111), (220), and (222) without any impurity and the formation of electrochemically inactive phase such as silicon carbide [26]. In addition, the intensities of Si 2p and O 1s peaks were reduced compared to that of C 1s peak due to the surface amorphous carbon coating layer of C@SGG, as shown in the XPS result of Fig. 2e.

3.2. Electrochemical characterization

The electrochemical properties of the SGG and C@SGG electrodes with an areal capacity of 3 mA h cm^{-2} were evaluated to demonstrate the beneficial effect of the uniform amorphous carbon coating layer through the coin half-cell test. Fig. 3a exhibits the voltage profiles of the electrode prepared with SGG and C@SGG at the first cycle with a current density of 0.3 mA cm^{-2} (0.1 C) in the voltage window of 0.01–1.5 V vs. Li/Li^+. The initial charging (delithiation) capacity of the SGG electrode

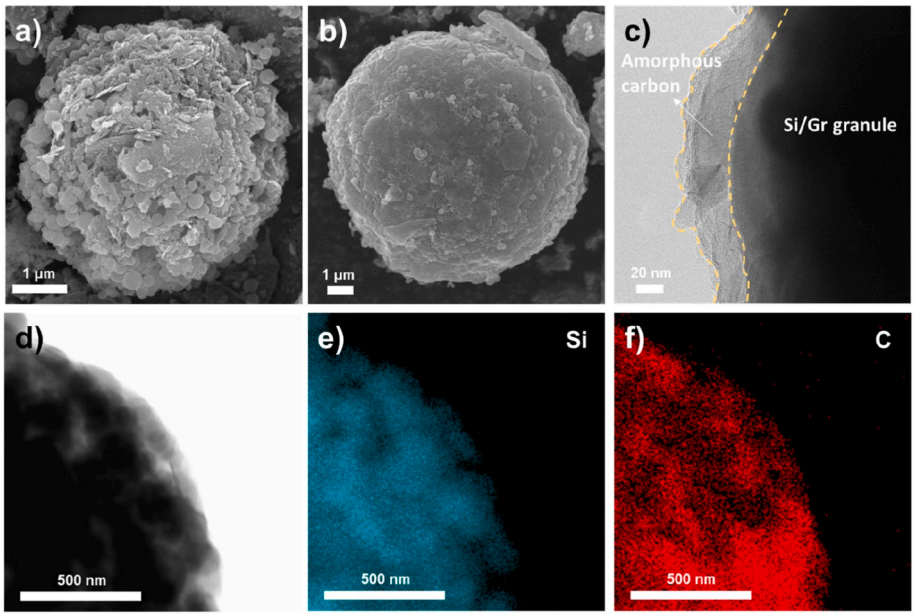


Fig. 1. SEM images of (a) SGG and (b) C@SGG. (c) TEM images of C@SGG to confirm the surface coating layer. TEM images of the (d) C@SGG and the corresponding EDS-mapping images (e, f).

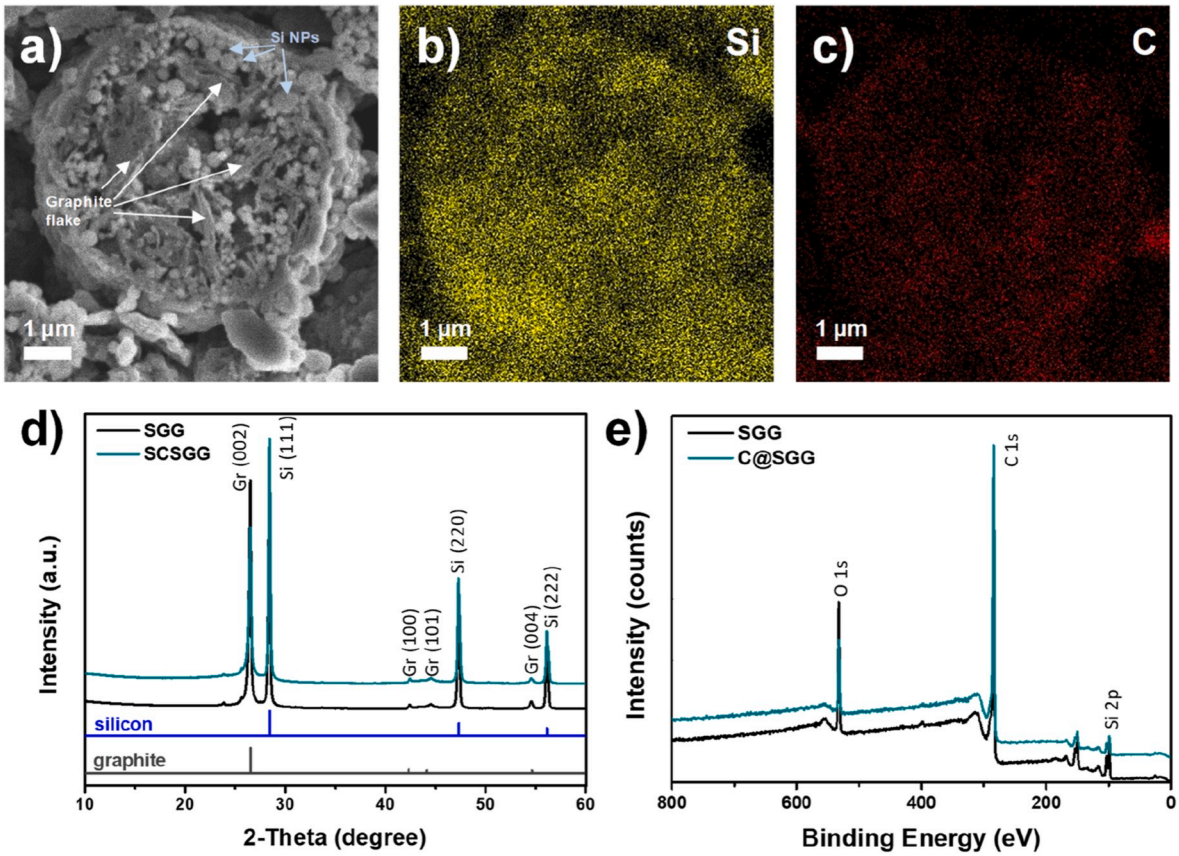


Fig. 2. (a) SEM image of the cross-section of C@SGG and its corresponding EDS mapping images (b) and (c). (d) The XRD diffraction patterns of SGG and C@SGG. (e) XPS spectra of SGG and C@SGG for full survey.

was 1650 mA h g^{-1} and its initial Coulombic efficiency was 79%, while the C@SGG electrode showed the initial charging capacity of 1150 mA h g^{-1} and the improved initial Coulombic efficiency of 85%. The uniform amorphous carbon coating layer of C@SGG could significantly reduce the surface area, which minimizes the formation of the solid electrolyte interphase (SEI) layer by blocking the direct contact between the organic electrolyte and Si/graphite active materials inside, resulting in the improved initial Coulombic efficiency. The initial charging capacities of the SGG and C@SGG electrodes were well matched with their theoretical capacities based on the content of Si analyzed by ICP-OES

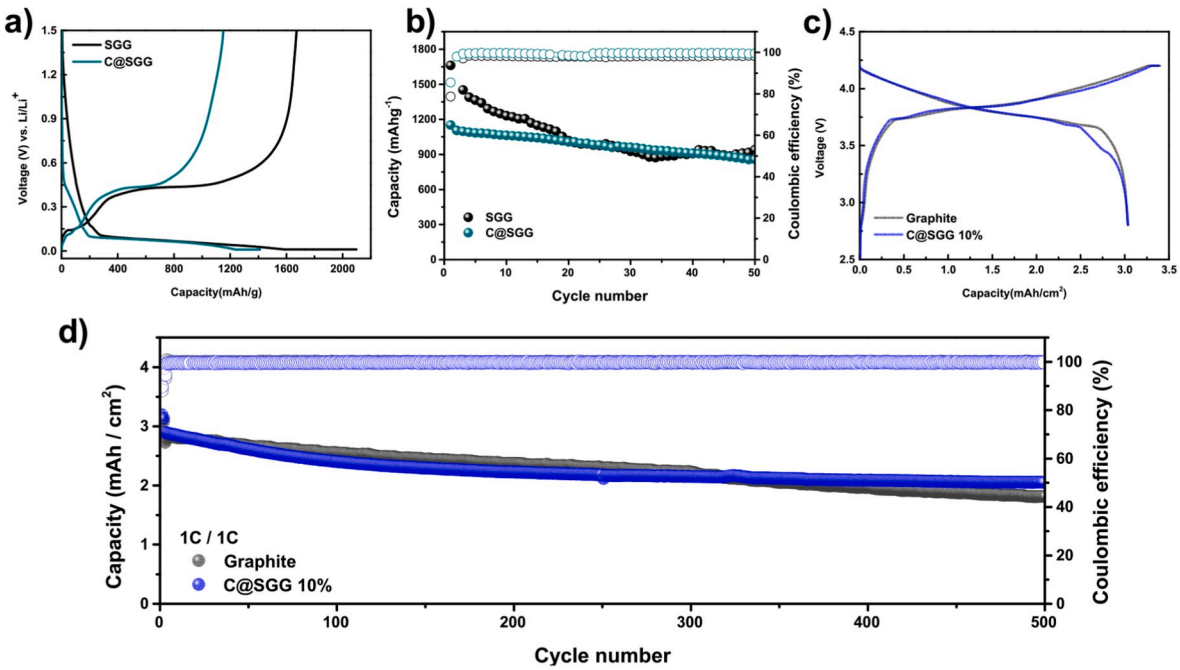


Fig. 3. (a) Initial charge-discharge profiles of the electrodes prepared with SGG and C@SGG at 0.1C rate between 0.01 and 1.5 V versus Li/Li$^+$. (b) Cycling performance of the electrodes prepared with SGG and C@SGG at 0.2 C rate. (c) Initial charge-discharge profiles of the full cells prepared with LCO/C@SGG with graphite at 0.1C rate between 2.8 and 4.2 V versus Li/Li$^+$. (d) Cycling performance of the full cells prepared with LCO/C@SGG with graphite at 1C rate (1C = 3.4 mA cm^{-2}).

measurement (Table S1). The SGG electrode exhibited the rapid capacity fading in earlier cycles and showed almost the same capacity as that of the C@SGG electrode after 20 cycles (Fig. 3b), which is attributed to the continuous electrolyte decomposition and the pulverization of the active materials during cycling. The C@SGG electrode showed the stable cycling performance at 0.2C rate over 50 cycles while the SGG electrode exhibited continuous capacity fading.

To investigate the practical feasibility, the C@SGG and graphite composite electrode was prepared for the pouch full cell evaluation using LCO as a cathode that has an areal capacity of 3.4 mA h cm^{-2} (Fig. S8) [27]. In order to achieve the high energy density of LIBs, the anode electrode was pressed to 1.5 g cc^{-1}. For C@SGG 10% electrode (C@SGG 10% and graphite 90%), both the volumetric and gravimetric energy density can be improved by about 19% compared to those of the graphite electrode. Fig. S9 exhibits 19% improvement in a specific capacity of the C@SGG 10% electrode compared to the conventional graphite electrode. Fig. 3c shows the initial voltage profiles of the full cells prepared with the graphite and C@SGG 10% electrodes at 0.1C (0.34 mA cm^{-2}) rate. The cell prepared with the C@SGG 10% electrode exhibited a high initial Coulombic efficiency of 90%, which is comparable to that of the conventional graphite electrode. To evaluate the long-term cycling performance, graphite and C@SGG 10% electrodes were evaluated at 1C/1C (charging/discharging) rate for 500 cycles, as shown in Fig. 3d. After two formation cycles at the initial 0.1 C and 0.2 C, the cell prepared with the C@SGG 10% electrode exhibited a capacity fading in earlier cycles and the capacity retention of 83% at 100 cycles from the 3rd cycle. However, the C@SGG 10% electrode showed the outstandingly stable cycling performance after 100 cycles due to the released stress, showing the capacity retention of 71% over 500 cycles. The graphite electrode showed more stable cycling performance at earlier cycles, showing the capacity retention of 93% at 100 cycles from the 3rd cycle. After that, the graphite electrode exhibited continuously faded capacities in the long-term cycling and had lower capacities than those of the C@SGG electrode after 300 cycles. The capacity retention of the graphite electrode was 66% over 500 cycles from the 3rd cycle. To investigate the composite electrode with a high content of C@SGG, the C@SGG 20% electrodes with different electrode densities of 1.2 and 1.5 g cc^{-1} were evaluated in the coin half-cell test. The C@SGG 20% electrode with an electrode density of 1.2 g cc^{-1} showed the high initial Coulombic efficiency of 90% and the reversible capacity (delithiation) of 510 mA h g^{-1} at 0.1 C compared to those of the C@SGG 20% electrode with an electrode density of 1.5 g cc^{-1}, as shown in Fig. S10. Furthermore, the C@SGG 20% electrode with 1.2 g cc^{-1} showed the stable cycling performance with the capacity retention of 80% at 0.2 C rate over 100 cycles. However, the cell prepared with the C@SGG 20% electrode with 1.5 g cc^{-1} exhibited poor cycling performance with the capacity retention of 63% at 0.2 C rate for 100 cycles. As the content of C@SGG in the electrode increases, more void space is needed inside the electrode that can mitigate the volume expansion of Si [28]. These results prove that even if we apply the Si/C composite material which has a high capacity, the microstructure of the electrode can have significant influence on the Coulombic efficiency and cycling performance of LIBs.

To evaluate the discharging rate capability of the pouch full cells, the discharging rate capability test was conducted up to 3.0 C discharging rate holding the charging rate of 0.2 C, as shown in Fig. 4a. The C@SGG 10% electrode showed the capacity retentions of 95.2, 90.8, 77.7 and 48.6%, respectively, at 0.5, 1.0, 2.0 and 3.0 C discharging rate. For the graphite electrode, the capacity retentions were 96.4, 92.9, 69.6 and 37.9%, respectively, at 0.5, 1.0, 2.0 and 3.0 C discharging rate. The lower capacity retention at a low C rate might be due to the capacity fading behavior of the C@SGG 10% electrode in earlier cycles [29]. However, the C@SGG 10% electrode showed much higher capacity retention at a high C rate, which substantiates the high power of the C@SGG and graphite composite electrode due to the high reactivity and Li diffusivity of nanoscale Si and the uniform amorphous carbon coating layer [30,31]. The fast charging test of the pouch full cells was conducted up to 3.0 C charging rate holding the discharging rate at 0.2 C as shown in Fig. 4b. The graphite and C@SGG 10% electrodes present no big difference in the capacity retentions up to 3.0 C because the cells were charged in the CC-CV (constant current and constant voltage) condition. However, the C@SGG 10% electrode showed higher charging capacities up to 3.0 C rate compared those of the graphite electrode when considering the charging capacity only in the constant current region (Fig. 4c and Fig. S11) [6]. Fig. 4d shows the electrochemical

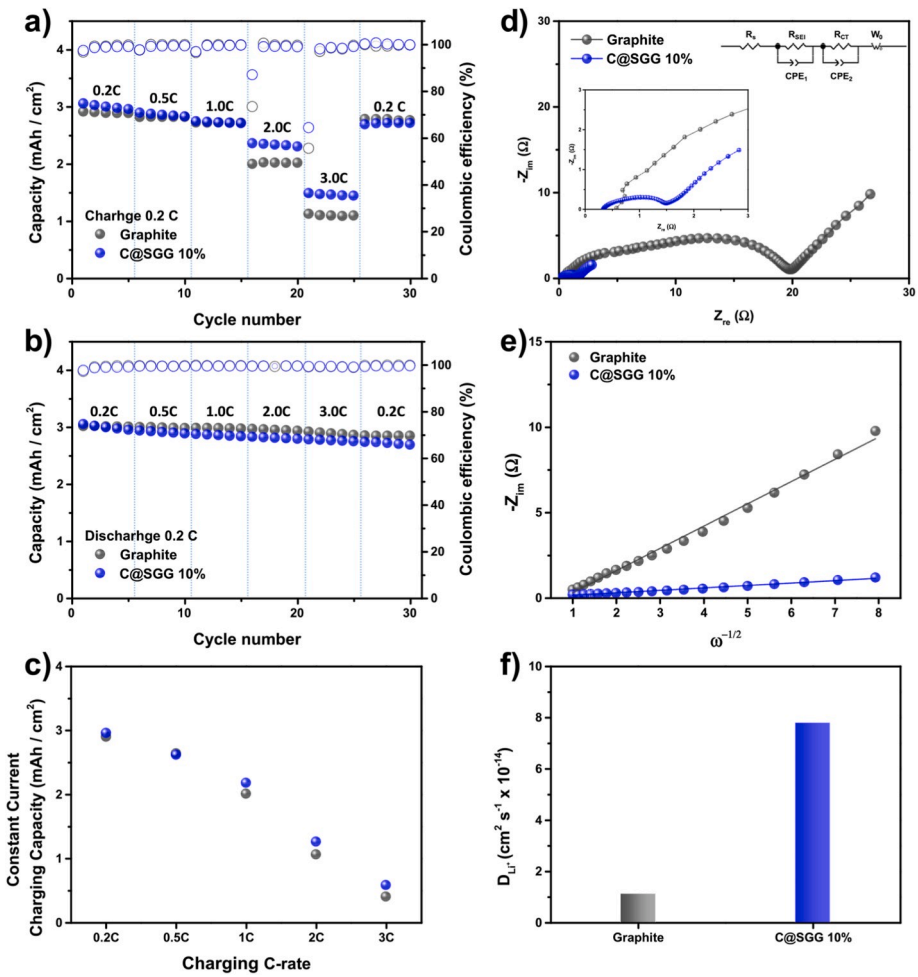


Fig. 4. Electrochemical characterization of the pouch full cells prepared with the graphite and C@SGG 10% electrodes. (a) Discharge rate capability test. (b) Charging rate capability test. (c) The constant current charging capacity of the full cells. (d) Nyquist plot of the full cells at 50% SOC (state of charge) after formation cycles. (e) Warburg plot of impedance as a function of $\omega^{-1/2}$ (angular frequency) in the low frequency range. (f) Li diffusion coefficient of the graphite and C@SGG 10% electrodes.

impedance spectra (EIS) of the pouch full cells prepared with the graphite and C@SGG 10% electrodes measured at 50% state of charge (SOC) after the two formation cycles, displayed in Fig. S12, at 0.1 C and 0.2 C [32,33]. The C@SGG 10% electrode showed the lower resistance of the SEI layer (R_{SEI}, the first semicircle in the high frequency range) and the lower charge transfer resistance (R_{ct}, the second semicircle in the medium frequency range) compared to those of the graphite electrode (Fig. 4d and Table 1). Furthermore, the C@SGG 10% electrode exhibited higher Li-ion diffusion coefficient (D_{Li^+}) of 7.803×10^{-14} compared to that of the graphite electrode (1.115×10^{-14}), as shown in Fig. 4e and f [34]. Even at 0% SOC, the C@SGG 10% electrode showed lower R_{ct} than graphite electrode, as shown in Fig. S13. The volume expansion of Si associated with Li inevitably leads to the stress on the graphite flake and the amorphous carbon coating layer in the composite electrode, however, the granular structure of the C@SGG enables to release the stress caused by the volume expansion of Si. Therefore, Li^+ ion of the shell of Li_xSi can efficiently move to the inner core of Si nanoparticle and graphite due to the released stress [35]. The above results demonstrate that the charging and discharging rate capabilities could be improved when employing C@SGG.

Table 1
Kinetic parameters of the graphite and C@SGG 10% electrodes in full cells.

Sample	R_s (Ω)	R_{SEI} (Ω)	R_{ct} (Ω)	D_{Li^+} ($cm^2\ s^{-1}$)
Graphite	0.571	5.013	16.814	1.115×10^{-14}
C@SGG 10%	0.345	0.605	1.035	7.803×10^{-14}

3.3. Electrode swelling characterization

In order to monitor the swelling behavior of the C@SGG and graphite composite electrode, we measured the thickness change of the pouch full cells using in-situ electrode thickness measurement system (TMS) (Fig. 5a) [36]. Generally, graphite is known to have a volume expansion of 10% when charged [37,38]. The graphite electrode exhibited the initial volume expansion of 12.7% when first charged at 0.1 C and the volume expansion of 14.5% when third charged at 1 C after two formation cycles at 0.1 C and 0.2 C. The graphite electrode showed the stable electrode swelling behavior during cycling, and the cycling expansion rate after 50 cycles at the discharged state is 5.6% based on the discharged state after two formation cycles. In case of the C@SGG 10% electrode, it showed volume expansions of 13.9% when first charged at 0.1C and 16.0% when third charged after two formation cycles. Furthermore, the C@SGG 10% electrode presented the saturated electrode swelling behavior during cycling, showing cycling expansion rate of 6.7% after 50 cycles. The C@SGG electrode showed 13.9% higher volumetric energy density compared to the graphite electrode even at a charged state. The stable swelling behavior of the C@SGG 10% electrode can not only allow the restricted space to be used freely in the practical cell design, but also improve the battery safety related to the electrode swelling issues. However, the C@SGG 20% electrode showed very high initial expansion of 23% when first charged and drastically increased electrode swelling behavior, which is due to the loss of the internal void space of the granule and the electrode caused by the roll pressing of the electrode [39]. The increased amount of C@SGG and the reduced buffering space in the electrode result in more stress to the active

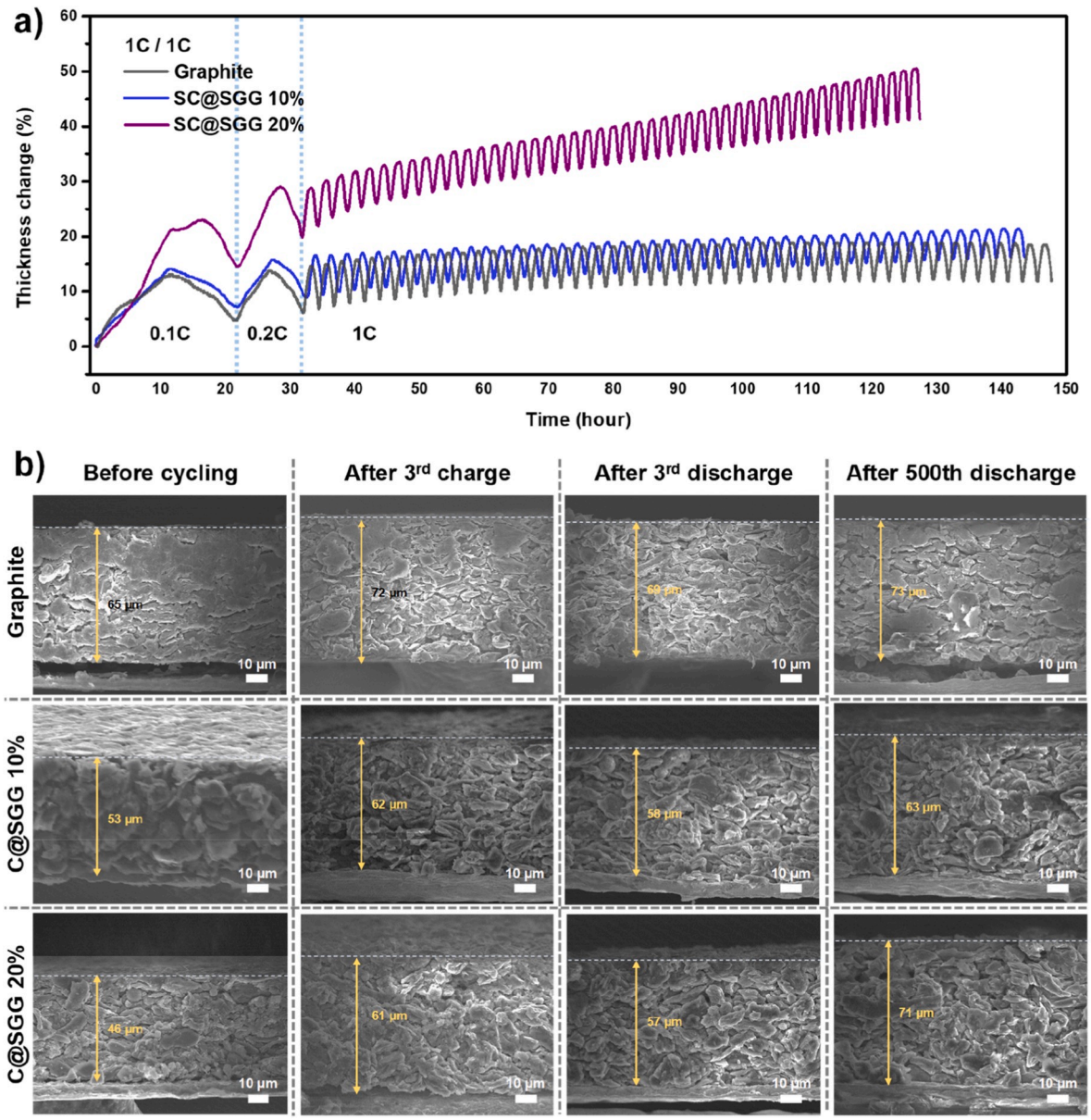


Fig. 5. In-situ electrode swelling behavior of the pouch full cells prepared with graphite, C@SGG 10%, and C@SGG 20% electrodes at 1 C (1C = 3.4 mA cm^{-2}) during 50 galvanostatic cycles. (b) Cross-sectional SEM images of graphite, C@SGG 10% and C@SGG 20% electrodes before and after cycling. The thickness change (%) was calculated based on the anode part.

materials and the electrode caused by the volume expansion of Si. In addition to the in-situ TMS evaluation, we investigated the cross-sectional SEM images of the composite electrodes during cycling, as shown in Fig. 5b. The graphite, C@SGG 10%, and C@SGG 20% electrodes showed the electrode thickness of 65, 53, and 46 μm, respectively before cycling, which is well matched with the electrode design with 3.4 mA h cm^{-2} and 1.5 g cc^{-1} (Table S2). Furthermore, the thickness of the composite electrodes is similar to the in-situ TMS results at the charged and discharged state at the 3rd cycle. The graphite electrode exhibited a cycling expansion of 6.2% and the C@SGG 10% electrode showed a cycling expansion of 9.4% at a discharged state after 500 cycles, which is comparable to the graphite electrode. The C@SGG 20% electrode exhibited the high cycling expansion of 30.4% at a discharged state after 500 cycles. Not only did the C@SGG 10% electrode initially show a low expansion rate, it also showed stable electrode swelling behavior over 500 cycles.

4. Conclusions

In conclusion, we developed C@SGG through the facile, low-cost, and scalable process and demonstrated the practical availability of C@SGG in terms of the electrochemical properties and the electrode swelling behavior. The C@SGG electrode exhibited the high reversible capacity of 1150 mA h g^{-1} and the improved initial Coulombic efficiency of 85% with a high areal capacity level of 3 mA h cm^{-2} in a coin half-cell test. The pouch full cell prepared with LCO as a cathode and the C@SGG 10% electrode as an anode showed the long-term cyclability with the capacity retention of 71% over 500 cycles at 1C. Furthermore, we elucidated the electrode swelling behavior of the C@SGG and graphite composite electrode through the in-situ TMS study. The C@SGG 10% electrode exhibited a low initial expansion of 16.0% at a charged state after the formation cycles and a stable and saturated cycling expansion of 6.7% over 50 cycles. Our findings suggest that C@SGG is a promising anode material that can be applied to the practical batteries with a stable cycling performance and safety with a

controlled swelling behavior for high energy density in LIBs.

Declaration of competing interest

The authors declare that there is no conflict of interest.

CRediT authorship contribution statement

Dongsoo Lee: Writing - original draft. **Seungcheol Myeong:** Investigation. **Insung Hwang:** Investigation. **Taeseup Song:** Funding acquisition, Project administration. **Makio Naito:** Funding acquisition, Project administration. **Ungyu Paik:** Supervision.

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Appendix A. Supplementary data

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